

QCD Chronometric Lock: All-Orders Threshold Propagation and the 2/27 Signal

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

18. Scale-invariant quantum definition

To prevent QCD from acquiring an independent subtraction standard, work in $d = 4 - 2\varepsilon$ and replace the external regularisation scale by

$$\mu_\chi = z \chi^{1/(1-\varepsilon)},$$

where z is dimensionless. The bare strong coupling is

$$g_{s,0} = \mu_\chi^\varepsilon Z_g g_s.$$

Renormalisation generates an infinite tower of scale-invariant higher-dimensional operators. The result is an all-orders Wilsonian EFT, not a finite-parameter UV completion. In a homogeneous background, every physical gap takes the form

$$M_A = f \mathcal{F}_A(g_i, z, c_k, \dots),$$

and

$$\Lambda_{n_f} = c_{n_f} \chi.$$

Slowly varying backgrounds receive derivative corrections such as

$$\Lambda_{n_f}(x) = c_{n_f} \chi(x) \left[1 + c_\square \frac{\square \chi}{\chi^3} + c_\partial \frac{(\nabla \chi)^2}{\chi^4} + \dots \right].$$

The physical statement is not merely “evaluate g_s at $\mu = \chi$.” It is that the quantum theory contains no independent dimensionful standard besides χ .

19. Exact threshold propagation theorem

Let a heavy coloured particle j with renormalisation-group-invariant mass $\widehat{m}_j$ be integrated out. Dimensional analysis gives

$$\Lambda_{n-1} = \widehat{m}_j \mathcal{F}_j \left(\frac{\Lambda_n}{\widehat{m}_j} \right).$$

Define

$$A_j = \frac{\partial \ln \mathcal{F}_j}{\partial \ln(\Lambda_n/\widehat{m}_j)}.$$

Differentiation yields

$$d \ln \Lambda_{n-1} = A_j d \ln \Lambda_n + (1 - A_j) d \ln \widehat{m}_j.$$

Define the lock and threshold defects

$$\delta_n = d \ln \frac{\Lambda_n}{\chi}, \quad \epsilon_j = d \ln \frac{\widehat{m}_j}{\chi}.$$

Then

$$\boxed{\delta_{n-1} = A_j \delta_n + (1 - A_j) \epsilon_j.}$$

For many thresholds,

$$\delta_3 = \left(\prod_{n=4}^N A_n \right) \delta_N + \sum_{n=4}^N (1 - A_n) \left(\prod_{k=4}^{n-1} A_k \right) \epsilon_n.$$

If the high-energy theory and every threshold are locked, the low-energy theory remains locked to all orders. Scheme and loop dependence live in the physical response coefficients A_j , not in the structure of the recursion [Chetyrkin1997].

20. A controlled coloured defect

Introduce one real scalar S and a vectorlike Dirac fermion

$$\Psi \sim (\mathbf{3}, \mathbf{1}, 0),$$

with

$$\mathcal{L}_{\Psi S} = \bar{\Psi}(i\gamma^\mu D_\mu - y_\chi \chi - y_S S)\Psi - \frac{1}{2}(\partial S)^2 - V_{\chi S},$$

$$V_{\chi S} = V_\chi(\chi) + \frac{\lambda_S}{4}(S^2 - r^2 \chi^2)^2 + \Delta V_{\text{ct}}.$$

At alignment, $S = r\chi$ and

$$M_\Psi = \chi(y_\chi + y_S r).$$

Parameterise the ratio fluctuation by

$$\frac{S}{\chi} = r e^\theta.$$

Then

$$d \ln \frac{M_\Psi}{\chi} = \varepsilon_\Psi d\theta, \quad \varepsilon_\Psi = \frac{y_S r}{y_\chi + y_S r}.$$

This is a physical crack in universal locking: it changes a dimensionless threshold ratio rather than merely rescaling every mass.

21. The 2/27 transmission coefficient

The threshold chain is

$$7 \xrightarrow{\Psi} 6 \xrightarrow{t} 5 \xrightarrow{b} 4 \xrightarrow{c} 3.$$

Assume the high-energy QCD scale and the top, bottom, and charm thresholds remain exactly locked. At one loop,

$$A_\Psi = \frac{19}{21}, \quad A_t = \frac{21}{23}, \quad A_b = \frac{23}{25}, \quad A_c = \frac{25}{27}.$$

Therefore

$$\begin{aligned} \mathcal{D}_{\Psi \rightarrow 3}^{(1)} &= \left(1 - \frac{19}{21}\right) \frac{21}{23} \frac{23}{25} \frac{25}{27} \\ &= \boxed{\frac{2}{27}}. \end{aligned}$$

The first controlled defect is

$$d \ln \frac{\Lambda_3}{\chi} = \frac{2}{27} \varepsilon_\Psi d\theta + O\left(\alpha_s \varepsilon_\Psi, \varepsilon_\Psi^2, \frac{\partial^2}{\chi^2}\right).$$

For a Dirac fermion in colour representation R ,

$$\mathcal{D}_{R \rightarrow 3}^{(1)} = \frac{4T(R)}{27}.$$

Again, the coefficient is not new. The new use is to make the first low-energy failure of universal time calibration calculable rather than arbitrary.

22. Clock and equivalence-principle predictions

A low-energy clock ratio obeys

$$\mathcal{S}_{AB} = \Delta K_\alpha d \ln \alpha + \Delta K_\mu d \ln \mu + \Delta K_q d \ln X_q + \dots,$$

with

$$\mu = \frac{m_p}{m_e}, \quad X_q = \frac{\widehat{m}}{\Lambda_3}.$$

In the pure-QCD benchmark,

$$d \ln \alpha = 0, \quad d \ln \mu \simeq \delta_3, \quad d \ln X_q = -\delta_3.$$

Thus

$$\mathcal{S}_{AB} = (\Delta K_\mu - \Delta K_q) \frac{2}{27} \varepsilon_\Psi d\theta + \dots$$

At leading sensitivity, the illustrative ratios give

$$\mathcal{S}_{\text{Sr/Cs}} = 2\mathcal{S}_{\text{Sr/CaF}}.$$

This is a rank-one pattern, conditional on one dominant scalar and the stated clock sensitivities [ClockData2023; ClockReview2025].

Let $a = f_a \theta$ and $\varphi = a/M_P$. The gluonic coupling is

$$d_g = \frac{\partial}{\partial \varphi} \ln \frac{\Lambda_3}{\chi} = \frac{2}{27} \varepsilon_\Psi \frac{M_P}{f_a} + \dots$$

In the one-scalar, unscreened, QCD-dominated limit, the anomalous clock redshift β_{AB} and Eötvös parameter η_{CD} satisfy

$$\frac{\beta_{AB}}{\eta_{CD}} = - \frac{\Delta K_\mu - \Delta K_q}{\Delta Q'_{\widehat{m}}{}^{CD}}.$$

For the project's leading Ti/Pt charge model, the conditional values are approximately 150.38 for Sr/CaF and 300.76 for Sr/Cs. They are model-consistency lines, not new experimental limits [DamourDonoghue2010; MICROSCOPE2022].

23. The radiative naturalness obstruction

The coloured fermion contributes

$$\Delta V_\Psi = -\frac{N_c}{16\pi^2} M_\Psi(\theta)^4 \left[\ln \frac{M_\Psi(\theta)^2}{\mu^2} - \frac{3}{2} \right].$$

The ratio mode receives

$$m_a^2 \sim \frac{N_c}{4\pi^2} \varepsilon_\Psi^2 \frac{M_\Psi^4}{f_a^2}.$$

A long-range observable signal is therefore catastrophically unnatural in the minimal one-threshold model. The programme requires a symmetry that suppresses the vacuum mass more strongly than the visible linear coupling.